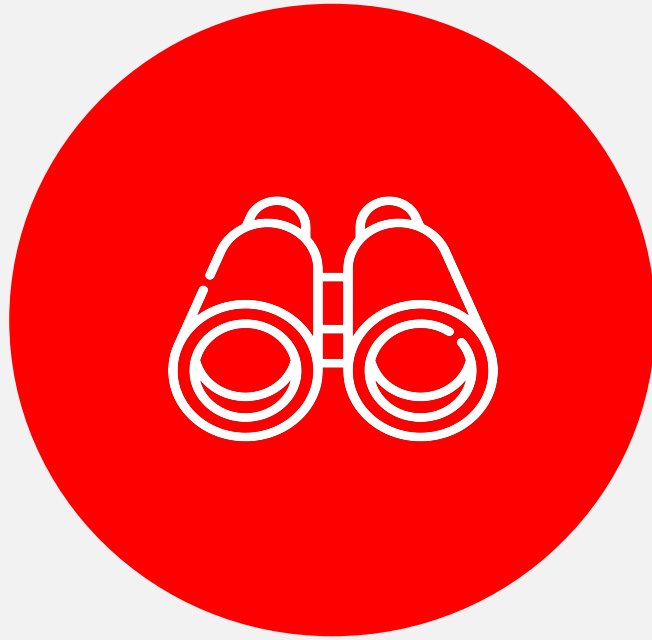




Initial Data Insights
Rossmann Pharmaceuticals Sales
Forecasting

Business Need

- Objective:
 - ✓ Forecast sales across all stores six weeks ahead to support the finance team
- Current Scenario:
 - ✓ Store managers rely on experience and judgment for predictions
- Importance:
 - ✓ Accurate predictions improve decision-making
 - ✓ Incorporate factors like promotions, holidays, seasonality, and locality

Data Overview

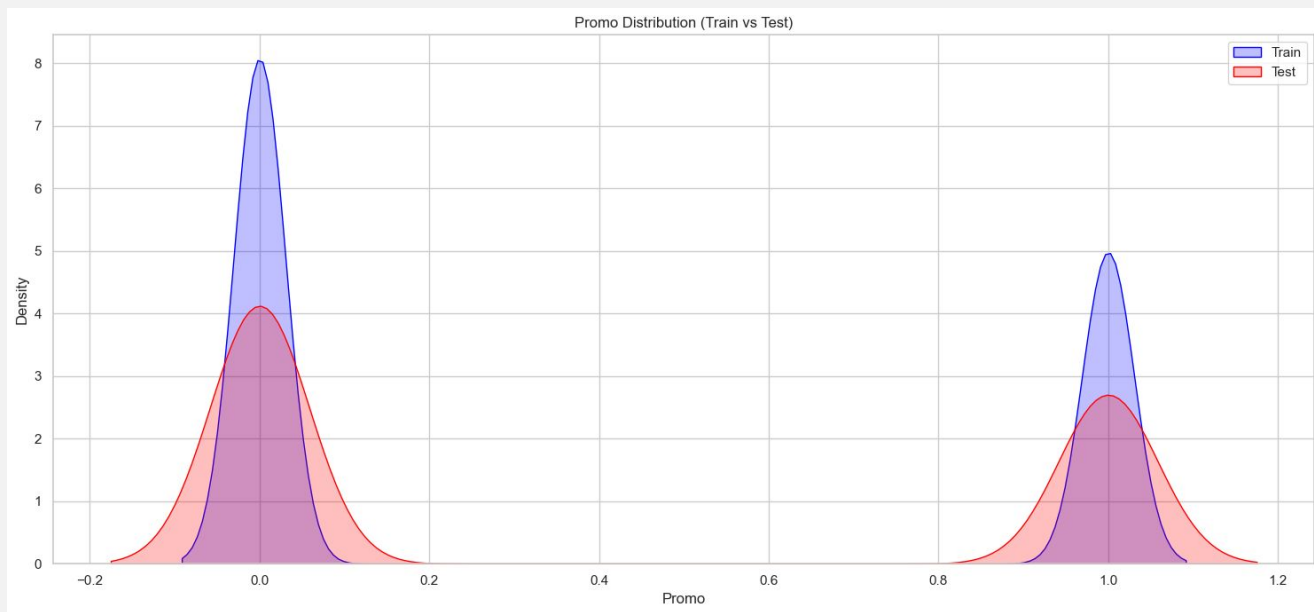
- Training Data: Historical sales records for stores
- Test Data: Records for prediction without sales and customers values
- Store Data: Information about stores
- Key Features:
 - ✓ 'Store': Unique identifier for each store
 - ✓ 'Sales': Target variable (daily turnover)
 - ✓ 'StateHoliday' & 'SchoolHoliday': Public event indicators
 - ✓ 'CompetitionDistance': Proximity of competitor stores
 - ✓ 'Promo' & 'Promo2': Promotional campaign indicators

Initial Data Cleaning

- Steps Taken:
 - Removed irrelevant 'Id' column.
 - Ensured correct 'StateHoliday' data type.
 - Unified 'Date' into datetime format.
- Missing Value Handling:
 - Applied 'mean' and 'median' strategies.
 - Logged all actions for reproducibility.

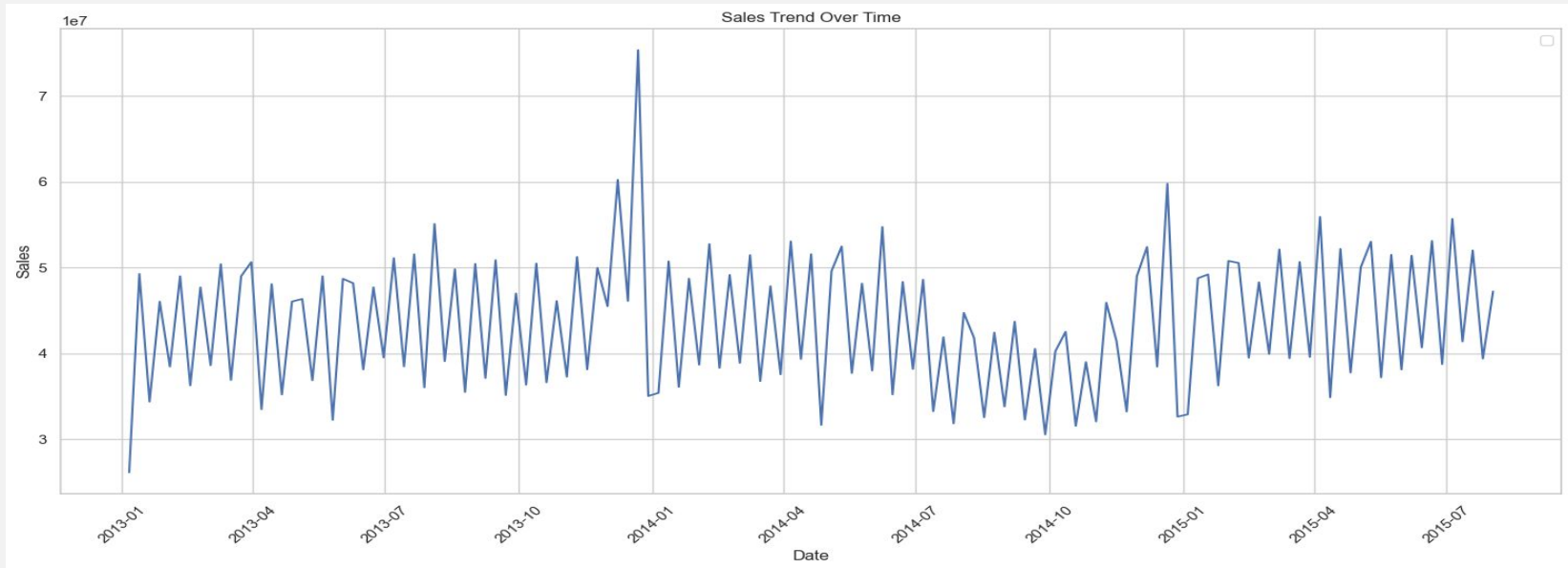
Distribution in Training and Test Sets

- Promo Analysis:
 - Distribution of promo occurrences aligns in training and test datasets.
 - Insight: Predictive models can generalize better due to similarity.



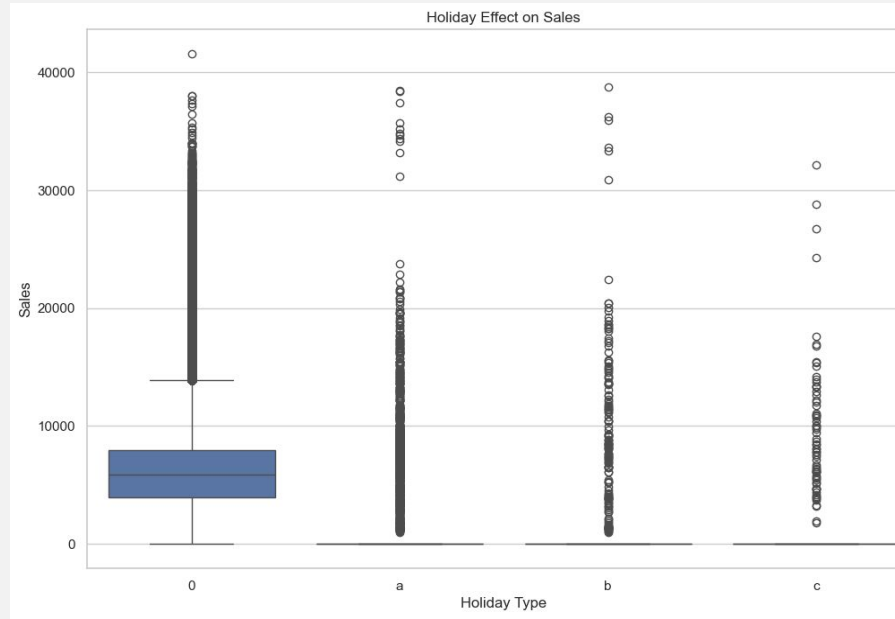
Cumulative Sales Over time

- Sales:
 - Constant sales overtime
- Insights:
 - Current strategies should be revised to increase sales



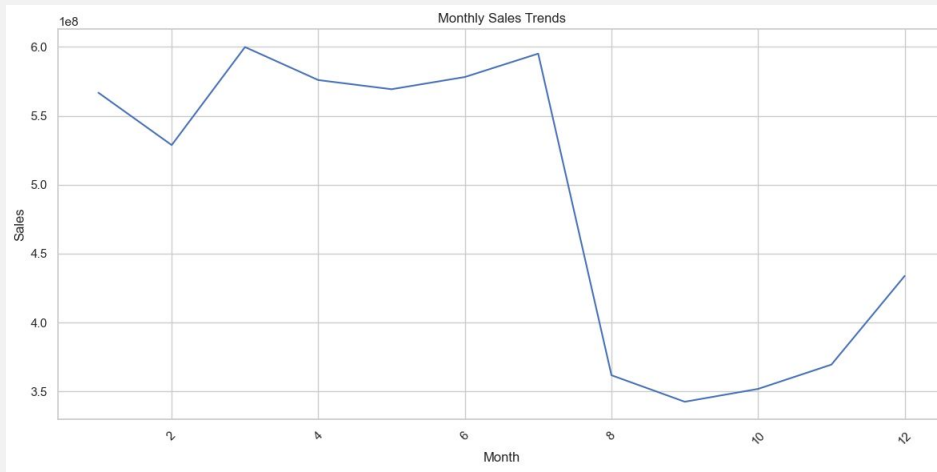
Sales Trends - During Holidays

- StateHoliday Impact:
 - Holidays show decline in average sales for all state holiday types
- Insights:
 - Needs further investigation based on each holiday for holiday-tailored strategies



Seasonal Analysis

- Seasonality Findings:
 - Sales rise from November to July and peak in March and end of June
 - Summer months display a steady rise due to vacation season.
- Recommendations:
 - Leverage peak seasons with targeted marketing.



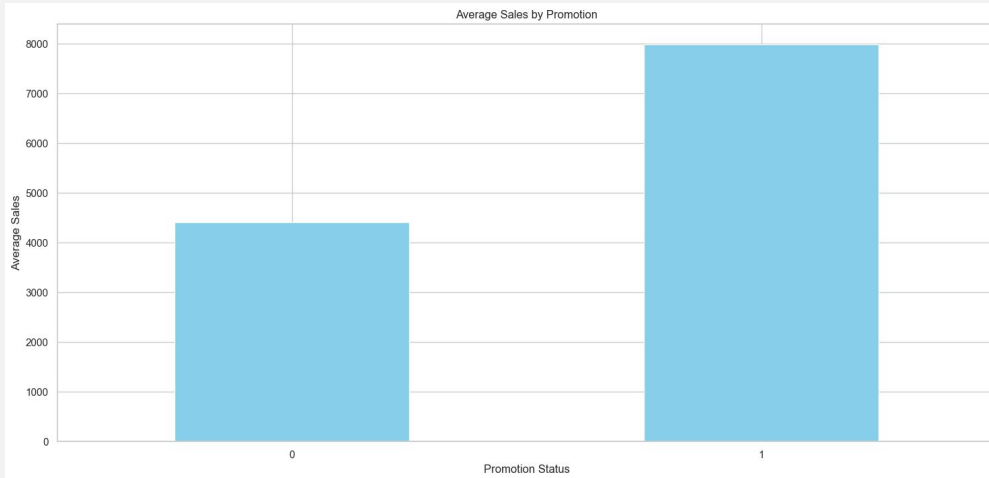
Sales and Customer Correlation

- Findings:
 - Strong correlation: More customers generally lead to higher sales
 - Store type 'd' drive fewer customers but higher sales per customer
- Insight:
 - Prioritize high-value promo targeting



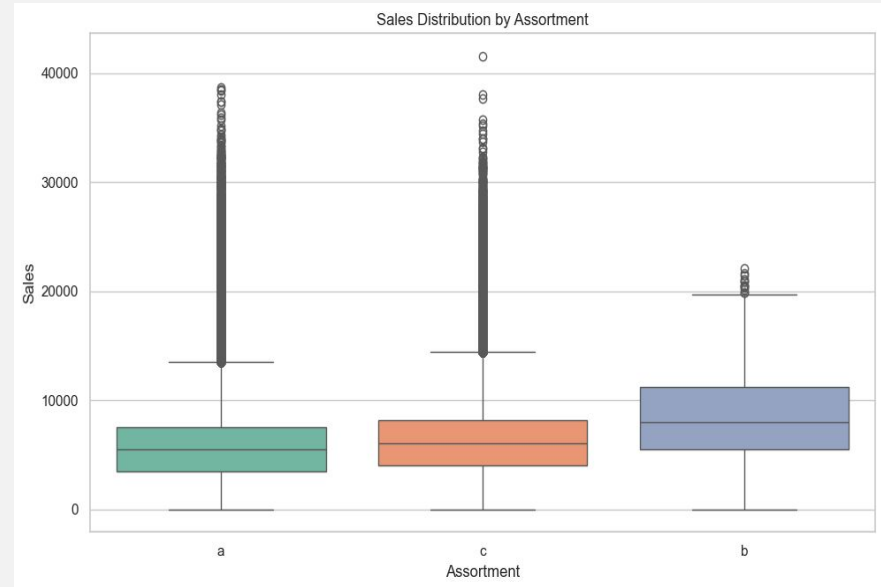
Impact of Promotions

- Promo Impact:
 - Promos significantly increase daily sales.
 - Higher customer turnout during promotional events.
- Optimizations:
 - Tailor promos by store types for optimal effectiveness.



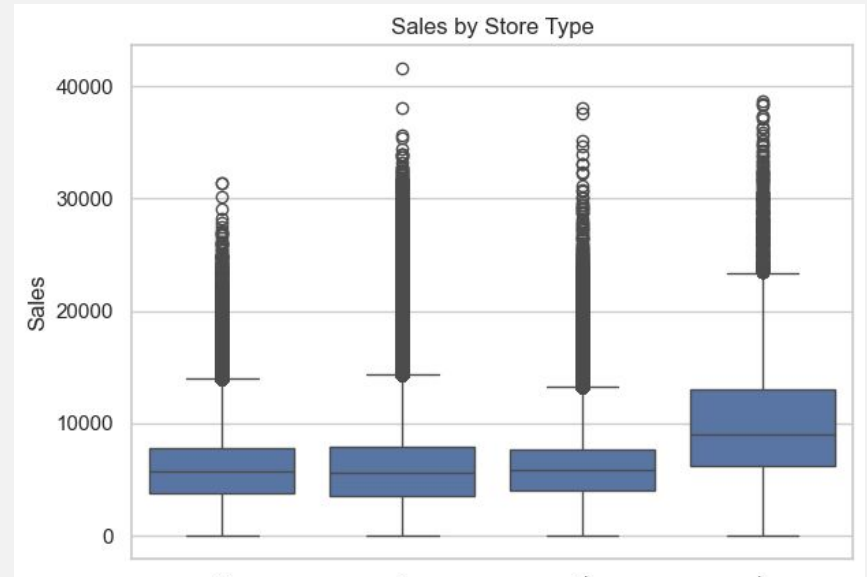
Assortment Types

- Insights:
 - Extended assortments ('c') outperform others in terms of sales.
 - Basic assortments ('a') are suitable for low-competition areas.
- Recommendation:
 - Extend assortments where feasible



Store Types

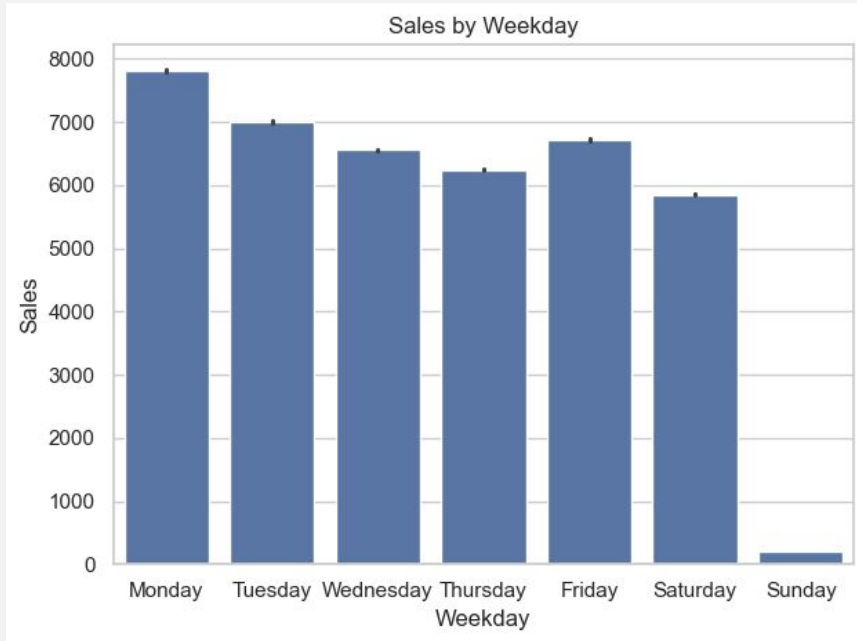
- Insights:
 - Store Type 'b' outperform others in terms of sales
 - Store Type 'a' shows comparatively less sales
- Recommendation:
 -



Store Hours and Behavior

- Insights:

- Sales are high on Friday and Saturday but almost none on Sunday as most Stores are closed

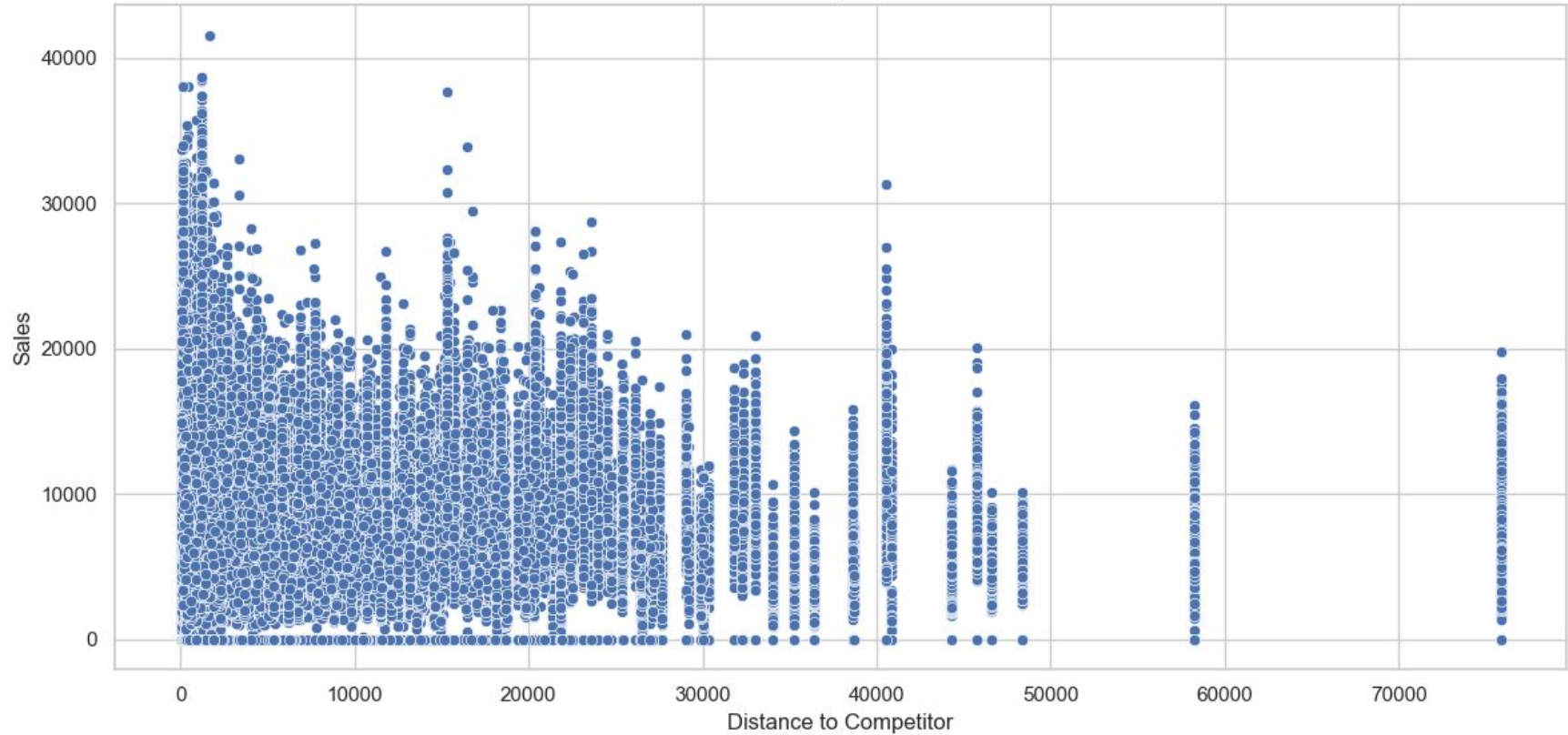


Competitor Proximity Analysis

- Findings:
 - Sales decline with decreasing competition distance
 - Nearby competitor openings reduce sales temporarily but effect diminishes over time
- Recommendation:
 - Focus competitive strategies in suburban and rural areas
 - Prepare marketing initiatives for anticipated competitor openings

Competitor Openings

Sales vs. Competitor Distance



Data Challenges and Log Integration

- Challenges:
 - Missing values in competitor distance and promo intervals.
 - Outliers in sales and customers.
- Log Integration:
 - Captured all processes (cleaning, missing value handling, outlier detection).

Key Insights Summary

- Promotions:
 - Essential for driving sales and customer engagement.
- Holidays and Seasonality:
 - Major contributors to sales variability.
- Store Characteristics:
 - Store type and proximity greatly affect sales.
- Customer Dynamics:
 - Sales directly linked to customer behavior patterns.

Recommendations

- Optimize promotions:
 - Focus on store-specific and customer-targeted campaigns.
- Enhance store hours:
 - Implement uniform operational schedules.
- Leverage seasonality:
 - Peak season marketing plans.
- Mitigate competition effects:
 - Proactive strategies near upcoming competitors.